

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B Tech-AI&DS	Date:	11-08-2026

Code of Conduct

I Y Greeshma Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y Greeshma Reddy

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

2. First-Order Predicate Calculus for Smart Manufacturing.

An Industry 4.0 manufacturing plant monitors machines using semantic rules.

Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

3. Word Sense Disambiguation in E-Commerce Search Engines.

An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand

"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Semantic Representation in Customer Support Chatbots

Understanding of Data

The chatbot uses semantic representations to convert customer queries into a structured meaning format consisting of an action, object, and entity.

Customer Query	Semantic Representation	Action	Object
Activate international roaming for my number.	ACTIVATE(Roaming, Customer)	ACTIVATE	Roaming
Deactivate caller tune service.	DEACTIVATE(CallerTune, Customer)	DEACTIVATE	Caller Tune
Check my data balance.	QUERY(DataBalance, Customer)	QUERY	Data Balance
Enable 5G service.	ACTIVATE(5GService, Customer)	ACTIVATE	5G Service

The chatbot logs contain four queries. Q1, Q3, and Q4 have correctly predicted intents, while **Q2** is incorrectly classified.

Analysis

1. Action-object relationships

- ACTIVATE + Roaming → The customer wants international roaming activated.
- DEACTIVATE + CallerTune → The customer wants the caller tune service deactivated.
- QUERY + DataBalance → The customer wants information about their current data balance.
- ACTIVATE + 5GService → The customer wants 5G service enabled.

The action is important because the same object can have different operations. For example, "Activate Caller Tune" and "Deactivate Caller Tune" refer to the same service but require completely different chatbot actions.

2. Semantic interpretation error

The error occurs in Q2:

- Actual Intent: Deactivate Caller Tune
- Predicted Intent: Activate Caller Tune

The object Caller Tune is correctly identified, but the action is incorrectly interpreted. The system has changed DEACTIVATE into ACTIVATE.

This is a significant semantic error because the chatbot could perform the opposite operation from what the customer requested.

3. Effect on decision accuracy

Meaning representation directly affects chatbot decision-making.

- Correct semantic representation → correct service/action.
- Incorrect action → incorrect service operation.
- Missing or ambiguous objects → wrong service selection.
- Incorrect negation such as "don't activate" can also cause serious errors.

For example, if `DEACTIVATE(CallerTune, Customer)` is incorrectly represented as `ACTIVATE(CallerTune, Customer)`, the chatbot may enable a service that the customer wanted to cancel.

Application of Concepts

The company can improve semantic understanding by using:

1. Intent classification to identify the customer's purpose.
2. Named Entity Recognition (NER) to identify services such as roaming, 5G, and caller tune.
3. Negation detection to distinguish "activate" from "do not activate" or "deactivate."
4. Context-aware models to understand previous messages.
5. Transformer-based NLP models for better semantic interpretation.
6. Intent validation before performing important customer-account actions.

A representation such as:

`DEACTIVATE(CallerTune, Customer)`

should be passed to the decision-making module rather than relying only on keyword matching.

Conclusion

Semantic representation provides a structured bridge between natural language and chatbot actions. In this case, Q2 is the only query with an incorrect semantic interpretation. Although the object was correctly identified, the action was reversed. Therefore, accurate action-object identification, negation handling, contextual analysis, and intent validation are essential for reliable telecom chatbots.

Clarity

The semantic representation clearly shows the relationship between customer, action, and service object. The example also demonstrates that even a small semantic error can result in a completely wrong customer-service operation.

2. First-Order Predicate Calculus for Smart Manufacturing

Understanding of Data

The manufacturing plant contains four machines:

- $M1 \rightarrow \text{Active}$
- $M2 \rightarrow \text{Active}$
- $M3 \rightarrow \text{Maintenance}$
- $M4 \rightarrow \text{Active}$

The predicate rules are:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

These rules allow the system to infer machine production status and product availability.

Analysis

1. Production data using First-Order Predicate Calculus

The production data can be represented as:

- $\text{Active}(M1)$
- $\text{Active}(M2)$
- $\text{Maintenance}(M3)$
- $\text{Active}(M4)$

Using:

$\text{Active}(x) \rightarrow \text{Producing}(x)$

we can infer:

- $\text{Producing}(M1)$
- $\text{Producing}(M2)$
- $\text{Producing}(M4)$

From:

$\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

we infer:

- $\neg \text{Producing}(M3)$

Therefore, M1, M2, and M4 are producing, while M3 is not producing.

2. Currently available products

The rule:

$\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$

means that a product is available if an active machine produces it.

However, the given production data does not specify which machine produces which product.

Therefore, the exact available products cannot be determined from the given information alone.

For example, if:

$\text{Produces}(\text{M1}, \text{Gear})$

and:

$\text{Active}(\text{M1})$

then:

$\text{Available}(\text{Gear})$

can be inferred.

3. Effect of maintenance on Gear production

M3 is under maintenance:

$\text{Maintenance}(\text{M3})$

Therefore:

$\neg \text{Producing}(\text{M3})$

If M3 is the machine responsible for producing Gear, then Gear production is affected and temporarily unavailable from M3.

However, without the fact:

$\text{Produces}(\text{M3}, \text{Gear})$

we cannot logically conclude that Gear production is affected.

Thus, predicate logic avoids making unsupported assumptions.

Application of Concepts

First-Order Predicate Calculus can be applied in manufacturing systems to represent:

- Machine status.
- Production activities.
- Product availability.
- Maintenance conditions.
- Machine-product relationships.
- Production failures.

For example:

$\text{Active}(\text{M1}) \wedge \text{Produces}(\text{M1}, \text{Gear}) \rightarrow \text{Available}(\text{Gear})$

and:

$\text{Maintenance}(\text{M1}) \rightarrow \neg \text{Producing}(\text{M1})$

can help an automated decision-support system determine whether production should continue.

Conclusion

First-Order Predicate Calculus provides a formal way to represent manufacturing knowledge and derive logical conclusions. M1, M2, and M4 are producing, while M3 is not producing because it is under maintenance. The availability of Gear specifically cannot be confirmed unless the machine-product relationship is provided. This makes predicate logic useful for reliable industrial decision-making.

Clarity

The use of predicates such as Active(x), Producing(x), and Maintenance(x) makes machine conditions easy to represent and reason about. It also prevents the system from drawing conclusions without sufficient facts.

3. Word Sense Disambiguation in E-Commerce Search Engines

Understanding of Data

Several search queries contain words with multiple meanings.

Query	Correct Sense	Evidence
Apple accessories	Technology Brand	iPhone Charger
Mouse wireless	Computer Device	Bluetooth Mouse
Java tutorial	Programming Language	Coding Lessons
Python course	Programming Language	Software Development Training

The clicked results provide strong contextual evidence for selecting the correct sense.

Analysis

1. Correct word sense

Apple accessories

Possible meanings:

- Fruit
- Technology brand

The clicked result is iPhone Charger, so "Apple" refers to the technology brand.

Mouse wireless

Possible meanings:

- Animal
- Computer device

The result is Bluetooth Mouse, so the intended meaning is a computer input device.

Java tutorial

Possible meanings:

- Island
- Programming language

The result is Coding Lessons, so Java refers to the programming language.

Python course

Possible meanings:

- Snake
- Programming language

The result is Software Development Training, so Python refers to the programming language.

2. Semantic cues

Important semantic cues include:

- "Accessories" → suggests a product or technology.
- "Wireless" → suggests a computer device.
- "Tutorial" → strongly suggests learning programming.
- "Course" → indicates educational content.
- Clicked results → provide direct behavioral evidence.
- Nearby words → help determine the intended meaning.

For example, "Python course" has a different meaning from "Python snake habitat" because the surrounding context changes the sense.

3. Impact of incorrect sense selection

Incorrect WSD can negatively affect an e-commerce search engine by:

- Showing irrelevant products.
- Reducing search accuracy.
- Decreasing customer satisfaction.
- Reducing click-through rate.
- Increasing search abandonment.
- Producing poor recommendations.

For example, if a customer searches for "Mouse wireless" and receives animal-related results, the search engine has failed to understand the customer's intention.

Application of Concepts

An industrial-scale WSD system can combine:

1. Contextual word embeddings using Transformer models.

2. Query expansion to understand related terms.
3. User browsing and purchase history.
4. Clicked-result information.
5. Product-category information.
6. Knowledge graphs connecting products and concepts.
7. Machine-learning-based intent classification.
8. Continuous feedback from user interactions.

For example:

Python + course + software → Programming Language

while:

Python + zoo + animal → Snake

This allows the search engine to select the appropriate sense based on context.

Conclusion

All four queries can be correctly disambiguated using contextual information. The clicked results strongly indicate the intended meanings. Accurate WSD is important because incorrect word-sense selection can lead to irrelevant search results and poor recommendations. A combination of contextual NLP models, behavioral data, knowledge graphs, and user feedback can significantly improve industrial-scale e-commerce search.

Clarity

The examples clearly demonstrate that the meaning of an ambiguous word depends on the words surrounding it and the user's search context. Therefore, WSD should consider the complete query rather than an individual keyword.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

Understanding of Data

The healthcare system processes sentences using syntactic structures such as:

Subject → Verb → Object

Examples:

- Doctor → prescribed → medicine
- Patient → reported → headache
- Nurse → monitored → patient
- Medicine → reduced → blood pressure

The system then assigns semantic roles such as Agent, Instrument, Recipient, and Symptom.

Analysis

1. Relationship between syntax and semantic roles

Syntactic structure helps determine the role played by each entity.

For example:

"Doctor prescribed medicine to patient."

- Doctor → Subject → Agent
- Medicine → Object → Theme/Medication
- Patient → Recipient

Similarly:

"Patient reported severe headache."

- Patient → Subject → Experiencer
- Headache → Object → Symptom

For:

"Nurse monitored patient continuously."

- Nurse → Agent
- Patient → Object/Entity being monitored

For:

"Medicine reduced blood pressure."

- Medicine → Cause/Agent-like entity
- Blood pressure → Affected condition/Target

2. Are the assigned semantic roles appropriate?

Some roles are appropriate, but there are areas that should be corrected.

Entity	Given Role	Evaluation
Doctor	Agent	Appropriate
Medicine	Instrument	Partially inappropriate
Patient	Recipient	Appropriate for "prescribed medicine to patient"
Headache	Symptom	Appropriate

The main issue is **Medicine → Instrument**.

In "Doctor prescribed medicine to patient," medicine is better represented as the **Theme/Medication**, rather than an Instrument.

Therefore, the semantic role assignment needs more detailed role labels.

3. Errors caused by incorrect parsing

Incorrect syntactic parsing can result in:

- Wrong doctor-patient relationships.
- Incorrect identification of symptoms.
- Wrong medication associations.
- Incorrect treatment information.
- Errors in medical information extraction.
- Incorrect clinical decision support.
- Misinterpretation of who performed an action.

For example, if the system incorrectly identifies "patient" as the Agent rather than the Recipient in "Doctor prescribed medicine to patient," it could create an incorrect medical record.

In healthcare, such errors can be more serious than ordinary text-processing errors because extracted information may influence clinical decisions.

Application of Concepts

Healthcare NLP systems can improve syntax-driven semantic analysis through:

1. Dependency parsing to identify relationships between words.
2. Semantic Role Labeling (SRL) to identify Agent, Patient, Theme, Experiencer, etc.
3. Medical Named Entity Recognition for diseases, medicines, symptoms, and people.
4. Medical-domain language models trained on clinical text.
5. Context analysis across multiple sentences.
6. Medical ontologies and knowledge graphs.
7. Human validation for critical medical information.
8. Rule-based validation to detect impossible or inconsistent relationships.

For example:

Doctor → prescribed → Medicine → to → Patient

can be represented semantically as:

Prescribe(Doctor, Medicine, Patient)

This structured representation is more useful for healthcare information systems.

Conclusion

Syntax-driven semantic analysis is important because grammatical structure helps determine the roles of entities in medical sentences. Most of the given roles are reasonable, but Medicine as an Instrument is not the most appropriate interpretation; it should generally be treated as the medication/theme of the prescribing event. Combining syntactic parsing, semantic role labeling, medical NER, contextual models, and domain knowledge can improve the accuracy of healthcare NLP systems.

Clarity

The relationship between syntax and semantics can be clearly understood through the **Subject–Verb–Object** structure. Accurate parsing ensures that doctors, patients, medicines, and symptoms are associated with the correct roles, which is essential for reliable healthcare information extraction.